

CHITOSE AP, Japan

WMO: 474340

Lat: 42.794N

Lon: 141.666E

Elev: 27

StdP: 101.01

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.0	(c) -15.8	(d) -20.8	(e) 0.6	(f) -17.1	(g) -18.2	(h) 0.8	(i) -14.3	(j) 12.9	(k) -0.5	(l) 11.4	(m) -1.0	(n) 1.2	(o) 210	(p) 0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.7	(c) 28.0	(d) 22.6	(e) 26.1	(f) 21.7	(g) 24.9	(h) 21.0	(i) 23.8	(j) 26.6	(k) 22.8	(l) 25.0	(m) 22.0	(n) 23.8	(o) 4.9	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.8	25.4	22.1	16.8	24.2	21.1	15.8	23.1	71.8	26.3	67.7	24.7	64.7	23.9	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.3	9.8	8.6	-21.3	30.6	2.4	1.6	-23.1	31.8	-24.5	32.7	-25.9	33.6	-27.6	34.8
			-21.6	25.1	2.4	1.0	-23.4	25.9	-24.8	26.5	-26.1	27.1	-27.8	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-6.8	-6.0	-1.1	5.6	11.4	15.4	19.6	21.1	17.4	10.4	3.4	-3.8
	DBStd	10.25	3.76	3.94	3.61	3.21	2.88	2.68	2.65	2.64	3.22	3.37	4.09	4.07
	HDD10.0	2138	521	449	345	137	19	1	0	0	0	37	201	428
	HDD18.3	4215	779	683	603	382	216	92	17	6	55	247	448	687
	CDD10.0	1146	0	0	0	5	61	164	297	345	222	49	3	0
	CDD18.3	181	0	0	0	0	0	6	56	93	27	0	0	0
	CDH23.3	677	0	0	0	0	6	25	197	359	89	0	0	0
	CDH26.7	110	0	0	0	0	1	3	29	70	7	0	0	0
(14) Wind	WSAvg	3.9	3.2	3.5	4.1	4.7	5.0	4.4	4.3	4.0	3.7	3.5	3.4	3.0
(15) Precipitation	PrecAvg	1233	75	63	57	73	108	91	144	165	166	104	98	90
	PrecMax	1569	110	141	101	132	187	255	302	337	277	150	206	168
	PrecMin	991	45	25	24	8	40	33	68	73	42	43	60	56
	PrecStd	141	15	26	20	35	38	44	54	69	65	29	34	27
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	5.1	10.9	19.2	23.8	26.0	29.2	31.0	27.8	21.2	16.2	9.9
		MCWB	2.7	2.8	6.4	11.7	14.7	18.9	23.4	24.7	22.3	16.1	12.7	8.6
	2%	DB	2.2	3.2	7.8	15.9	20.9	23.2	27.2	28.8	25.8	19.8	14.1	6.9
		MCWB	0.9	1.5	4.6	9.7	13.9	17.8	22.4	23.3	20.9	15.4	11.4	5.0
	5%	DB	1.1	2.1	5.9	13.8	19.0	22.0	25.8	26.9	24.1	18.0	12.2	4.2
		MCWB	-0.2	0.6	3.5	8.7	12.6	17.1	21.5	22.5	20.1	14.2	9.8	2.4
	10%	DB	-0.1	1.0	4.2	11.2	17.0	20.1	23.9	25.1	22.9	16.8	10.8	2.2
		MCWB	-1.3	-0.4	2.2	7.1	11.9	16.3	20.6	21.6	19.3	13.8	8.5	0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.5	6.9	12.3	16.3	20.2	24.6	25.5	23.7	18.1	14.0	8.7
		MCDB	3.5	4.2	9.1	17.1	21.2	23.9	27.9	29.6	26.1	20.0	15.4	9.5
	2%	WB	1.1	1.9	5.2	10.6	14.9	18.8	23.3	24.2	22.5	16.4	12.2	5.1
		MCDB	1.9	2.9	6.9	14.3	18.9	22.1	26.1	27.2	24.4	18.5	13.8	6.3
	5%	WB	0.0	0.7	3.7	9.4	13.8	17.7	22.2	23.4	21.5	15.3	10.5	2.6
		MCDB	0.7	1.8	5.0	12.4	17.3	20.7	24.5	25.6	23.0	17.3	12.0	3.9
	10%	WB	-1.6	-0.3	2.5	8.0	12.9	16.8	21.2	22.5	20.1	14.0	8.8	0.9
		MCDB	-0.4	0.9	4.1	10.4	15.9	19.4	23.1	24.4	22.0	16.2	10.4	2.1
(35) Mean Daily Temperature Range	MDBR		10.1	10.4	8.5	9.8	9.6	7.9	6.7	6.7	9.0	10.7	9.3	9.3
	5% DB	MCDBR	9.4	9.7	9.5	13.5	13.2	10.8	9.5	8.8	9.4	11.4	11.5	9.9
		MCWBR	8.6	8.5	7.2	8.0	7.1	5.9	5.2	4.5	5.5	7.8	9.3	8.5
	5% WB	MCDBR	9.3	9.3	8.8	11.6	10.6	9.1	7.9	7.3	7.1	9.6	10.7	9.7
(40) Clear-Sky Solar Irradiance	taub		0.340	0.392	0.466	0.526	0.545	0.526	0.512	0.480	0.421	0.407	0.388	0.356
	taud		2.036	1.889	1.788	1.782	1.834	1.962	2.069	2.174	2.295	2.191	2.139	2.052
	Ebn at Noon		751	766	756	747	748	765	771	783	806	756	696	689
	Edh at Noon		111	153	193	210	204	181	161	140	116	113	101	101
(44) All-Sky Solar Radiation	RadAvg		1.52	2.32	3.49	4.58	5.11	4.96	4.50	4.26	3.69	2.70	1.62	1.23
	RadStd		0.10	0.17	0.25	0.47	0.44	0.41	0.49	0.32	0.33	0.15	0.14	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	-107	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page